

Text Mining



Aritra Sen
Data Scientist
Cognizant Tech Solutions
Kolkata

What is Text Mining

- *“It may be loosely characterized as the process of analyzing text to extract information that is useful for particular purposes”*

Ian H Witten

*Computer Science, University of Waikato, Hamilton,
New Zealand*

- *The phrase “text mining” is generally used to denote any system that analyzes large quantities of natural language text and detects lexical or linguistic usage patterns in an attempt to extract probably useful (although only probably correct) information*

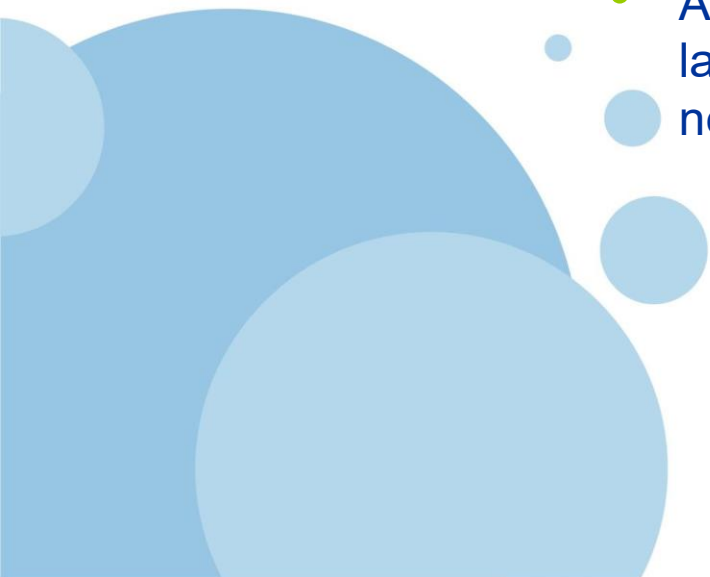
- **Sebastiani, F. (2002)** “Machine learning in automated text categorization.” *ACM Computing Surveys*, Vol. 34, No. 1, pp. 1–47

Do we need text mining?

We do need a process which can read and redirect us to those documents which are of prime interests...!!

- Before answering to this question let me ask a few more questions
 - How much time do you need to read a novel like “Gone with the Wind”? A computer with i5 processor can read it within a minute
 - How many web pages can you see and read in a day? A computer program can crawl through thousands of pages a day (depending on the speed of internet connections)
 - How much time do you need to check whether two documents are similar or not? A computer can check similarities among more than 100 documents within 2 minutes
 - How much time do you need to see what the world is twitting about Obama? A computer can give a rough idea in 5 minutes

Can text mining replace a human reader?

- The answer is still “NO” as of now
 - A computer cannot understand the plot of the novel “Gone with the Wind”
 - A computer will not understand whether “*the dog bites the boy*” or “*the boy bites the dog.*”
 - A computer program can crawl over thousands of pages but it may fail to understand the link between the pages
 - A computer cannot understand “Hinglish” or sms languages which are used quite frequently in social networking sites
- 

Then why do we need text mining?

- It can help in understanding whether a novel belongs to suspense thriller or love story
- It helps in capturing plagiarism
- It can scan through mail servers and thousands of mails to identify threats
- It can go through RSS feeds to capture important news and give automated alerts
- It can read thousands of online reviews and blogs to see how topic of discussions have changed over time (say related to Obama)

Text Analysis concept

- Analyzing text is a difficult task because of linguistic patterns which vary considerably across different languages
- However, occurrence of words in relation to other words does give an understanding of the subject as a whole (irrespective of the grammar).
- For example: “A dog had bitten the boy. The boy was taken to hospital. He was given tetanus injection”... In this entire paragraph “dog” has appeared once. The word “boy” has appeared twice, “bitten”, “hospital”, “tetanus” and “injection” has appeared only once. Hence, even if the entire paragraph is not given, we can have some idea what would have happened if these words are given to us.

Text Mining

- It is a relatively new concept and lots of researches are going on to extract specific information from unstructured text documents.
- Text mining involves breaking the texts into either words, or sentences or paragraphs
- When the texts are broken down into words, it is called “bag of words” concept. In this concept the linguistic patterns of a text is lost and hence, so far, **text mining cannot be considered as a substitute of a human reader. But it acts as an aid in choosing what to read thoroughly**

Text Mining

- In “bag of words” concept each unique word acts as a variable and each document acts as a case or respondent
- Texts are broken using “regular expression” or “regex” which can recognize occurrence of characters and it also splits the words based on some particular pattern. In most of the cases, two words are separated using a ‘whitespace’ character and text miners start collecting the words on the basis of splitting the entire texts using these ‘whitespace’ delimiters. This process is called “Tokenization”
- After tokenization, further filtering of texts are done
- Stopwords are removed
- All texts are converted to lower case
- Stemming is done for identifying root words
- As required, POS tagging and token-filtering are also done

Apollo
apollo
APOLLO
apollo

market
↓
root

marketing
↓
market

marketed
↓
market

Example

- “**Sambalpur University** will install special **software** to ensure proper verification of the originality of **PhD thesis** during registration of doctorate degrees. This is being done to check allegation of **plagiarism** in **PhD thesis** paper, said vice-chancellor of the **Sambalpur University** Prof B C **Barik** on Thursday. He said there were allegations against one research scholar. "We have decided to install special **software** to ensure authenticity of **PhD thesis** paper. This is a step to end **plagiarism**," said **Barik**. He, however, said talk is on to finalize which **software** would be perfect for the work.”
- This text contains 61 different words
- Some of the important words are highlighted
- If these highlighted words are given, we can have some idea that some software is there in Sambalpur University which can check PhD thesis for plagiarism

- Tokenization produces series of unique words which act as variables and after tokenization, an $N \times M$ matrix is created where N is the number of documents and M is the number of unique words appeared in at least one of the documents.
- This matrix is popularly known as “Document Term Matrix”. This matrix acts as the basic input data for further analysis
- However, different measures can be taken to building such matrix. The very basic one is “Binary Term Occurrence Matrix”

Example

- Doc1: Ram has a Dog and a Cat
- Doc2: Ravi has a Cat
- Doc3: Shivam has a Parrot
- Doc4: He has a Rabbit

These 4 docs have 10 different words

If based on term occurrences a document term matrix is created, it will look like



Doc	Cat	He	Parrot	Rabbit	Ram	Ravi	Shivam	a	dog	has	and
1	1	0	0	0	1	0	0	2	1	1	1
2	1	0	0	0	0	1	0	1	0	1	0
3	0	0	1	0	0	0	1	1	0	1	0
4	0	1	0	1	0	0	0	1	0	1	0

freq of term
has - 4
dog - 1
a - 5

- “Frequency of Term Occurrence Matrix” and “Tf-Idf Matrix” can also be produced for other analyses
- Frequency of Term Occurrence Matrix contains the frequency of occurrence of a single word in different documents (as shown in the previous example)
- Tf-Idf Matrix is one of the most popular matrix because the Tf-Idf value signifies importance of a word in a specific document with respect to other words in the same document.

TF-IDF (Term Frequency-Inverse Document Frequency)

- Tf-Idf values are calculated for each word in each document.
- It is a simple multiplication of Term Frequency of a word with inverse of document frequency.
- Mathematically it is written as

$$TF - IDF = TF_{ij} \times \log \frac{N}{n_i}$$

- TF_{ij} is the term frequency of word i in the j^{th} document, n_i is the total number of documents containing the word i and N is the total number of documents

D	C	H	P	R	R	R	S		d	h	a
oc	at	e	rot	bbi	a	av	Shi		o	a	n
1	1	0	0	0	1	0	0	2	1	1	1
1	1	0	0	0	1	0	0	2	1	1	1
2	1	0	0	0	0	1	0	1	0	1	0
3	0	0	1	0	0	0	1	1	0	1	0
4	0	1	0	1	0	0	0	1	0	1	0

Normalized Term Frequency

- TF calculated is normalized by dividing the term frequency by the square root of sum of the squares of the frequency of all the words occurring in that document
- Mathematically it given by:

$$TF_{i,d} = \frac{w_i}{\sqrt{\sum_{i=1}^n w_i^2}}$$

Where w_i is the frequency of i th word which is belonging to the document d and the document contains n words

Example: Calculation of TF-IDF

Normalized
TF-IDF = 0.5×1.3862

- Let us look at the previous example
 - Doc1: Ram has a Dog
 - Doc2: Ravi has a Cat
 - Doc3: Shivam has a Parrot
 - Doc4: He has a Rabbit
- In Doc1 “Ram” is appearing once and there are 4 words in it with each word having frequency of occurrence 1
- Hence normalized TF_{Ram} is given by

$$TF_{Ram} = \frac{1}{\sqrt{1^2 + 1^2 + 1^2 + 1^2}} = 0.5$$

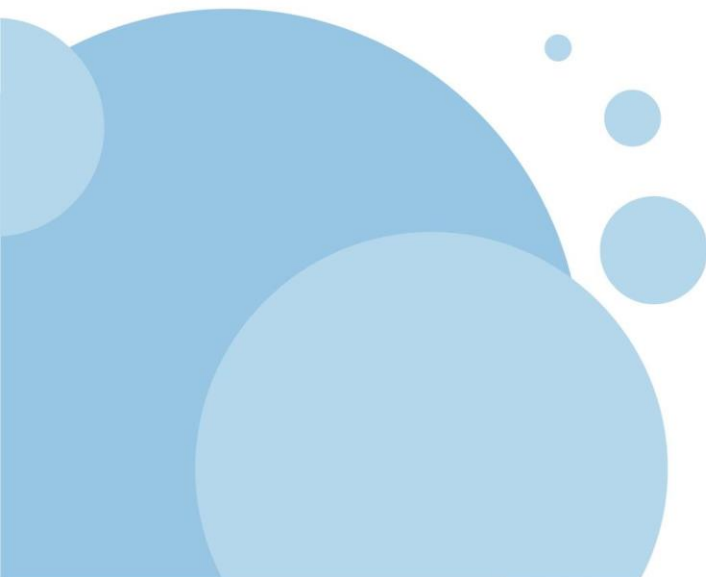
- Since “Ram” is appearing in only one document and there are 4 documents available, IDF_{Ram} is given by

$$IDF_{Ram} = \log \frac{4}{1} = 1.3862$$

- Hence non-normalized TF-IDF of Ram is $1 \times 1.3862 = 1.3862$

Important concepts in Text Mining

- Information Retrieval
 - *“It is finding the documents which contains answers to the question but not the answer itself”*
- Natural Language Processing
 - *“To achieve better understanding of natural language by use of computers ”*
- Information Extraction
 - *“Deals with extraction of specific information from text documents ”*



Data Mining in Texts

- Classification
 - KNN
 - Naïve Bayesian classification
 - SVM
 - Decision Tree
 - Linear Classification
- Clustering
 - K-Mean
 - Hierarchical clustering
 - Topic modeling
 - Semi-supervised clustering
- Information Extraction
 - Hidden Markov Model
 - Conditional Random Field

- Typically falls into one of two categories
 - Analysis of text: I have a bunch of text I am interested in, tell me something about it
 - E.g. sentiment analysis, “buzz” searches
 - Retrieval: There is a large corpus of text documents, and I want the one closest to a specified query
 - E.g. web search, library catalogs, legal and medical precedent studies

Text Mining: Analysis

- Which words are most present
- Which words are most surprising
- Which words help *define* the document
- What are the interesting text phrases?

Text Mining: Retrieval

- Find k objects in the corpus of documents which are most similar to my query.
- Can be viewed as “interactive” data mining - query not specified a priori.
- Main problems of text retrieval:
 - What does “similar” mean?
 - How do I know if I have the right documents?
 - How can I incorporate user feedback?

Text Retrieval: Challenges

- Calculating similarity is not obvious - what is the distance between two sentences or queries?
- Evaluating retrieval is hard: what is the “right” answer ? (no ground truth)
- User can query things you have not seen before e.g. misspelled, foreign, new terms.
- Goal (score function) is different than in classification/regression: not looking to model all of the data, just get best results for a given user.
- Words can hide semantic content
 - Synonymy: A keyword T does not appear anywhere in the document, even though the document is closely related to T , e.g., data mining
 - Polysemy: The same keyword may mean different things in different contexts, e.g., mining